

```
pip install nltk
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)  
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.4.2)  
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)  
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.  
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.3)
```

```
import nltk, warnings  
warnings.filterwarnings('ignore')  
for pkg in ['punkt_tab', 'stopwords', 'wordnet', 'movie_reviews']:  
    nltk.download(pkg, quiet=True)  
print('NLP Environment Ready!')
```

```
NLP Environment Ready!
```

```
corpus=["My Name is Muskan.",  
        "I am a Data Science Student.",  
        "I am currently in 5th Semester.",  
        "I am pursuing. my Data Science degree from K.R. Mangalam University.",  
        "My University is based in Sohna, Gurugram.",]  
with open('corpus.txt', 'w') as f:  
    f.write('\n'.join(corpus))  
print('corpus.txt written:', len(corpus), 'documents')
```

```
corpus.txt written: 5 documents
```

```
from nltk.tokenize import word_tokenize, sent_tokenize  
docs=open('corpus.txt').read().splitlines()  
text=' '.join(docs)  
sents=sent_tokenize(text)  
words=word_tokenize(text)  
print('Documents:', len(docs))  
print('Sentences:', len(sents))  
print('Words (tokens):', len(words))  
print('Unique tokens:', len(set(w.lower() for w in words)))
```

```
print('Unique tokens: ', len(set(words)), 'for all docs',)
print('Average word per document:', round(len(words)/len(docs), 1))
print("token",words )
```

Documents: 5

Sentences: 7

Words (tokens): 42

Unique tokens: 25

Average word per document: 8.4

token ['My', 'Name', 'is', 'Muskan', '.', 'I', 'am', 'a', 'Data', 'Science', 'Student', '.', 'I', 'am', 'cur

sents

```
['My Name is Muskan.',
 'I am a Data Science Student.',
 'I am currently in 5th Semester.',
 'I am pursuing my Data Science degree from K.R.',
 'Mangalam University.',
 'My University is based in Sohna, Gurugram.']
```

docs

```
['My Name is Muskan',
 'I am a Data Science Student',
 'I am currently in 5th Semester',
 'I am pursuing my Data Science degree from K.R. Mangalam University',
 'My University is based in Sohna, Gurugram']
```

```
sample=docs[3]
print('Raw text: ', sample)
print('Word tokens:', word_tokenize(sample))
print('Sentence tokens:', sample.split()[:8], '...')
```

Raw text: I am pursuing . my Data Science degree from K.R. Mangalam University.

Word tokens: ['I', 'am', 'pursuing', '.', 'my', 'Data', 'Science', 'degree', 'from', 'K.R', '.', 'Mangalam',

Sentence tokens: ['I', 'am', 'pursuing', '.', 'my', 'Data', 'Science', 'degree'] ...